Question 1: Recovery Time Objective (RTO) Calculation

A company has determined that its maximum acceptable downtime for a critical system is 4 hours. During a recent disaster recovery test, it was observed that the system could be restored in 5.5 hours. Calculate the difference between the actual recovery time and the Recovery Time Objective (RTO).

Solution:

Actual Recovery Time = 5.5 hours

RTO = 4 hours

Difference = Actual Recovery Time - RTO = 5.5 hours - 4 hours = 1.5 hours

The actual recovery time exceeds the RTO by 1.5 hours.

Question 2: Annualized Loss Expectancy (ALE) Calculation

An organization estimates that a specific threat could lead to a data center outage costing $250,000 per incident. If the likelihood of this event occurring is once every 10 years, calculate the Annualized Loss Expectancy (ALE).

Solution:

Single Loss Expectancy (SLE) = $250,000

Annual Rate of Occurrence (ARO) = 1/10 = 0.1

ALE = SLE × ARO = $250,000 × 0.1 = $25,000

The ALE for this threat is $25,000.

Question 3: Data Backup Storage Requirement

A data center needs to back up 15 terabytes (TB) of data weekly. If the backup storage system has a data compression ratio of 2:1, calculate the amount of storage required to store four weeks of backups.

Solution:

Uncompressed weekly backup size = 15 TB

Compression ratio = 2:1

Compressed weekly backup size = 15 TB / 2 = 7.5 TB

Storage required for four weeks = 7.5 TB/week × 4 weeks = 30 TB

The storage system needs 30 TB to store four weeks of compressed backups.

Question 4: Power Consumption and UPS Capacity

A data center operates equipment that consumes a total of 120 kilowatts (kW). The uninterruptible power supply (UPS) system has a capacity of 600 kilowatt-hours (kWh). Calculate how long the UPS can sustain the data center's operations during a power outage.

Solution:

Total power consumption = 120 kW

UPS capacity = 600 kWh

Duration = UPS capacity / Total power consumption = 600 kWh / 120 kW = 5 hours

The UPS can sustain operations for 5 hours during a power outage.

Question 5: Bandwidth Requirement for Data Replication

A data center needs to replicate 500 gigabytes (GB) of data to a remote site daily. If the replication must be completed within a 10-hour window each day, calculate the minimum required bandwidth in megabits per second (Mbps).

Solution:

Total data to replicate = 500 GB = 500 × 1024 MB = 512,000 MB

Replication window = 10 hours = 10 × 3600 seconds = 36,000 seconds

Required bandwidth in megabytes per second (MBps) = Total data / Replication window = 512,000 MB / 36,000 seconds ≈ 14.22 MBps

Since 1 byte = 8 bits, required bandwidth in Mbps = 14.22 MBps × 8 ≈ 113.76 Mbps

The minimum required bandwidth for data replication is approximately 113.76 Mbps.